

UP LT GRADE • COMPUTER SCIENCE MAINS

Internet & Web Technology — High-Probability Numerical & Practical Questions with Solutions

Selected for Likelihood of Appearing, Based on the Confirmed Exam Pattern

Coverage: IP addressing & subnetting, URL parsing, HTTP status codes, bandwidth/timing calculations, HTML/JavaScript code tracing, and port number recall.

Each question is tagged with its appearance likelihood, with the full step-by-step method shown — this subject's "numericals" are mostly applied/analytical rather than pure arithmetic.

Practice set • step-by-step solutions • exam-style presentation

Table of Contents

Q1. IP Address Classification	Page 3
Q2. Subnetting Calculation	Page 4
Q3. URL Structure Parsing	Page 5
Q4. HTTP Status Code Categorization	Page 6
Q5. File Download Time Calculation	Page 7
Q6. HTML Table Rendering (rowspan/colspan) Trace	Page 8
Q7. JavaScript Code Output Tracing	Page 9
Q8. Well-Known Port Numbers	Page 10
Q9. Transmission Time vs Propagation Delay	Page 11
Q10. Web Page Size & Load Time (Bits/Bytes Conversion)	Page 12

Q1. IP Address Classification

High Chance

Question

For the IP address 172.16.5.4, identify (a) its Class, (b) the Default Subnet Mask, and (c) the Network and Host portions.

Step 1: Identify the Class from the First Octet

Classful Ranges (First Octet)

Class A: 1–126 Class B: 128–191 Class C: 192–223 Class D: 224–239 (Multicast) Class E: 240–255 (Experimental)

First octet = 172, which falls in the range 128–191 → **Class B**

Step 2: Default Subnet Mask for Class B

Class B uses the first 2 octets for Network, last 2 for Host → Default Mask = **255.255.0.0 (/16)**

Step 3: Split Network and Host Portions

Network Portion = **172.16** Host Portion = **5.4**

Final Answer

Class B, Default Mask = 255.255.0.0, Network = 172.16.x.x, Host = x.x.5.4

Q2. Subnetting Calculation

Moderate Chance

Question

Given the network 192.168.10.0/26, find (a) the subnet mask, (b) the number of usable hosts per subnet, and (c) the Network Address and Broadcast Address of the THIRD subnet block.

Step 1: Subnet Mask and Block Size

/26 means 26 network bits, 6 host bits.

Subnet Mask = 255.255.255.192 Block Size = $2^6 - 192 = 64$

Step 2: Usable Hosts per Subnet

Formula

Usable Hosts = $2^{(\text{host bits})} - 2$ (subtract Network Address and Broadcast Address)

Usable Hosts = $2^6 - 2 = 64 - 2 = 62$ hosts

Step 3: List Subnet Blocks (multiples of 64) and Find the 3rd

Subnet #	Range
1	192.168.10.0 - .63
2	192.168.10.64 - .127
3	192.168.10.128 - .191
4	192.168.10.192 - .255

Final Answer

Subnet Mask = 255.255.255.192, Usable Hosts/Subnet = 62

3rd Subnet: Network Address = **192.168.10.128**, Broadcast Address = **192.168.10.191** (usable range: .129 to .190)

Q3. URL Structure Parsing

High Chance

Question

Parse the following URL into its individual components:

`https://www.example.com:8080/products/view.php?id=25&cat=books#reviews`

Break Down Each Component

Component	Value	Meaning
Protocol	https	Secure HTTP — encrypted communication
Host/Domain	www.example.com	Identifies the server
Port	8080	Non-default port the server is listening on (default HTTPS port is 443)
Path	/products/view.php	Specific resource/script being requested on the server
Query String	id=25&cat=books	Two parameters passed to the script: id=25 and cat=books
Fragment	#reviews	Jumps to a specific section within the loaded page (handled by the browser, not sent to the server)

Final Answer

All 6 components identified above — Protocol, Host, Port, Path, Query String, and Fragment together make up the complete URL.

Q4. HTTP Status Code Categorization

High Chance

Question

For each HTTP status code below, identify its category and exact meaning: 200, 301, 404, 500.

HTTP Status Code Categories

Range	Category
1xx	Informational
2xx	Success
3xx	Redirection
4xx	Client Error
5xx	Server Error

Applying to the Given Codes

Code	Category	Meaning
200	Success	OK — request succeeded
301	Redirection	Moved Permanently — resource has a new URL
404	Client Error	Not Found — requested resource doesn't exist
500	Server Error	Internal Server Error — something failed on the server side

Final Answer

200 = Success (OK), 301 = Redirection (Moved Permanently), 404 = Client Error (Not Found), 500 = Server Error (Internal Server Error)

Q5. File Download Time Calculation

Moderate Chance

Question

A file of size 15 MB needs to be downloaded over a connection with bandwidth 2 Mbps. Calculate the download time.

Step 1: Convert File Size to Megabits

Key Conversion

1 Byte = 8 bits, so MB (Megabytes) $\times$ 8 = Mb (Megabits)

File Size = 15 MB $\times$ 8 = **120 Megabits**

Step 2: Apply Download Time Formula

Formula

Download Time = File Size (in bits) / Bandwidth (in bits per second)

Download Time = 120 Mb / 2 Mbps = **60 seconds**

Final Answer

Download Time = **60 seconds (1 minute)**

Common Mistake to Avoid

Bandwidth is measured in bits per second (Mbps), but file sizes are usually given in Bytes (MB) — always convert to the SAME unit (bits) before dividing, or the answer will be off by a factor of 8.

Q6. HTML Table Rendering (rowspan) Trace

Moderate Chance

Question

For the following HTML table code, determine the visual grid size (rows × columns) and explain how the rowspan affects rendering:

```
<table border="1">
  <tr><td rowspan="2">A</td><td>B</td><td>C</td></tr>
  <tr><td>D</td><td>E</td></tr>
</table>
```

Step 1: Count Declared Cells Per Row

Row 1: 3 <td> tags — A (rowspan=2), B, C

Row 2: 2 <td> tags — D, E (only 2, NOT 3, because cell A already occupies a slot in Row 2 due to rowspan)

Step 2: Reconstruct the Visual Grid

A (spans both rows)	B	C
↓ (A continues here)	D	E

Final Answer

Visual Grid = **2 rows × 3 columns** (6 total grid positions). Only 5 <td> elements are written in the code, because cell A's rowspan="2" makes it occupy 2 of the 6 grid positions using just ONE <td> declaration.

Q7. JavaScript Code Output Tracing

High Chance

Question

Trace the output of the following JavaScript code:

```
var x = 5;  
var y = "10";  
document.write(x + y);  
document.write("<br>");  
document.write(x + Number(y));
```

Step 1: Analyze the First Statement (x + y)

x = 5 (Number), y = "10" (String).

When a Number is added to a String using +, JavaScript performs **String Concatenation**, not arithmetic addition.

Result: "5" + "10" = "510" (a string)

Step 2: Analyze the Second Statement (x + Number(y))

Number(y) explicitly CONVERTS the string "10" into the number 10.

Now both operands are numbers: 5 + 10 = 15 (arithmetic addition)

Final Answer

Output: 510

15

This demonstrates JavaScript's "type coercion" — the + operator behaves differently (concatenation vs addition) depending on operand types, a very commonly tested JS quirk.

Q8. Well-Known Port Numbers

Very High Chance

Question

Match each protocol to its default (well-known) port number: HTTP, HTTPS, FTP, SMTP, POP3, Telnet, DNS.

Answer Table

Protocol	Default Port
HTTP	80
HTTPS	443
FTP	21 (control) / 20 (data)
SMTP	25
POP3	110
Telnet	23
DNS	53

Final Answer

See table above — these 7 port numbers are the most frequently tested across all networking/web technology descriptive papers.

Why This Is Very High-Probability

Port number recall is one of the easiest, most direct-recall questions examiners can ask — expect it standalone OR embedded inside a larger protocol-explanation question. Memorizing this exact table is high-value for very little effort.

Q9. Transmission Time vs Propagation Delay

Moderate Chance

Question

A sender transmits 1,000,000 bits of data over a link with bandwidth 1 Mbps. The link is 2000 km long, and the signal propagates at 2×10^8 m/s. Calculate (a) Transmission Time, (b) Propagation Delay, and (c) Total Time.

(a) Transmission Time

Formula

Transmission Time = Data Size (bits) / Bandwidth (bits per second)

Transmission Time = $1,000,000 / 1,000,000 = 1$ second

(b) Propagation Delay

Formula

Propagation Delay = Distance / Propagation Speed

Distance = 2000 km = 2,000,000 m

Propagation Delay = $2,000,000 / (2 \times 10^8) = 0.01$ seconds (10 ms)

(c) Total Time

Total Time = Transmission Time + Propagation Delay = $1 + 0.01 = 1.01$ seconds

Final Answer

Transmission Time = 1 s, Propagation Delay = 10 ms, Total Time = **1.01 seconds**

Q10. Web Page Size & Load Time

Moderate Chance

Question

A web page consists of 3 images of 250 KB each, plus HTML+CSS totaling 50 KB. Calculate (a) the total page size in Megabits, and (b) the estimated load time over a 4 Mbps connection.

(a) Total Page Size

Images: $3 \times 250 \text{ KB} = 750 \text{ KB}$

HTML+CSS: 50 KB

Total = $750 + 50 = 800 \text{ KB}$

Convert to bits: $800 \text{ KB} \times 8 = 6400 \text{ Kilobits} = \mathbf{6.4 \text{ Megabits}}$

(b) Estimated Load Time

Formula

Load Time = Total Page Size (Megabits) / Bandwidth (Mbps)

Load Time = $6.4 \text{ Mb} / 4 \text{ Mbps} = \mathbf{1.6 \text{ seconds}}$

Final Answer

Total Page Size = 6.4 Megabits (800 KB), Estimated Load Time = **1.6 seconds**

Exam Connection

This exact style of question connects directly to the "Web Page Design Principles" topic (Load Speed) — expect it phrased as a design/optimization justification question as often as a pure calculation.

— End of Practice Set —

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